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Register No : 231701007 **Name: Badrish R S**

A python program to implement boosting

Aim:

To implement boosting using the AdaBoost algorithm on the given dataset for classification.

Requirements:

Jupyter Notebook with Python.

PROGRAM:

```
import pandas as pd import numpy as np from

sklearn.ensemble import AdaBoostClassifier from

sklearn.tree import DecisionTreeClassifier from

sklearn.metrics import accuracy_score from

sklearn.model_selection import train_test_split

import matplotlib.pyplot as plt


# Load dataset df = pd.read_csv('/mnt/data/gym

statistics(2).csv')


print("Dataset Preview:") print(df.head())


# Convert categorical data to numeric  df_encoded
= pd.get_dummies(df)
```

```
# Split features and target (last column as target)

X = df_encoded.iloc[:, :-1] y = df_encoded.iloc[:,

-1]


print("\nFeature shape:", X.shape) print("Target

shape:", y.shape)


# Train-test split (IMPORTANT)

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)


# Initialize AdaBoost with Decision Tree (stump) base_model

= DecisionTreeClassifier(max_depth=1)


model = AdaBoostClassifier(

estimator=base_model,     n_estimators=50,

learning_rate=1.0

)


# Train model model.fit(X_train, y_train)

# Predictions y_pred =

model.predict(X_test)


print("\nPredictions:") print(y_pred)
```

Accuracy accuracy =

accuracy_score(y_test, y_pred)

print("\nAccuracy:", accuracy)

Feature Importance Plot importances =

model.feature_importances_

plt.figure(figsize=(10,6)) plt.barh(X.columns,

importances) plt.title("Feature Importance

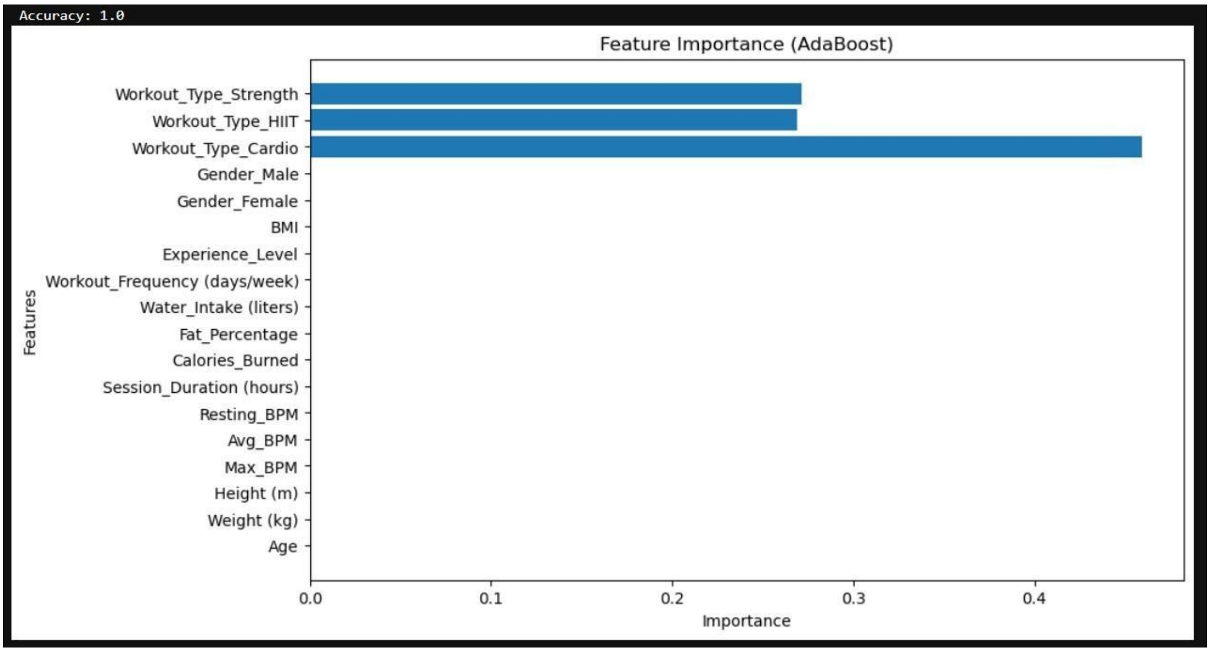
(AdaBoost)") plt.xlabel("Importance")

plt.ylabel("Features") plt.show()

Output:

Feature shape: (199, 18)

Target shape: (199,)



Result:

The AdaBoost model was successfully trained and achieved improved classification accuracy on the dataset.